

Tech & Teach

Digital Chip Design Program

RTL Design

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Gate-level

Task #1.1

```
module task1_1(input logic x1,  
input logic x2,  
input logic x3,  
output logic f);
```

```
    logic nx2, w1, w2;
```

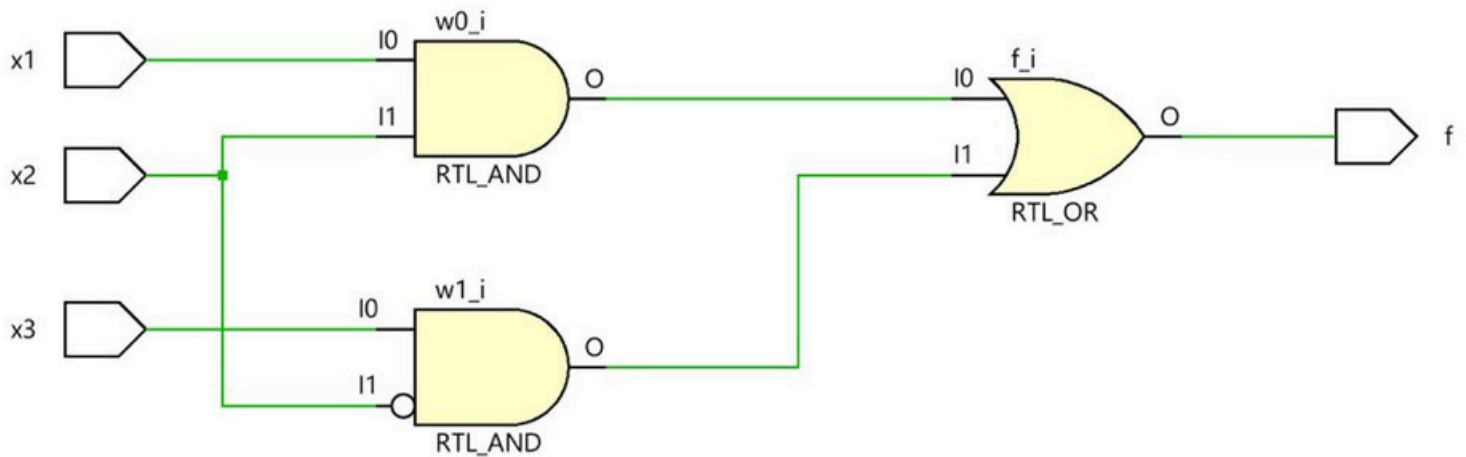
```
    not n1(nx2, x2);
```

```
    and a1(w1, x1, x2);
```

```
    and a2(w2, x3, nx2);
```

```
    or o1(f, w1, w2);
```

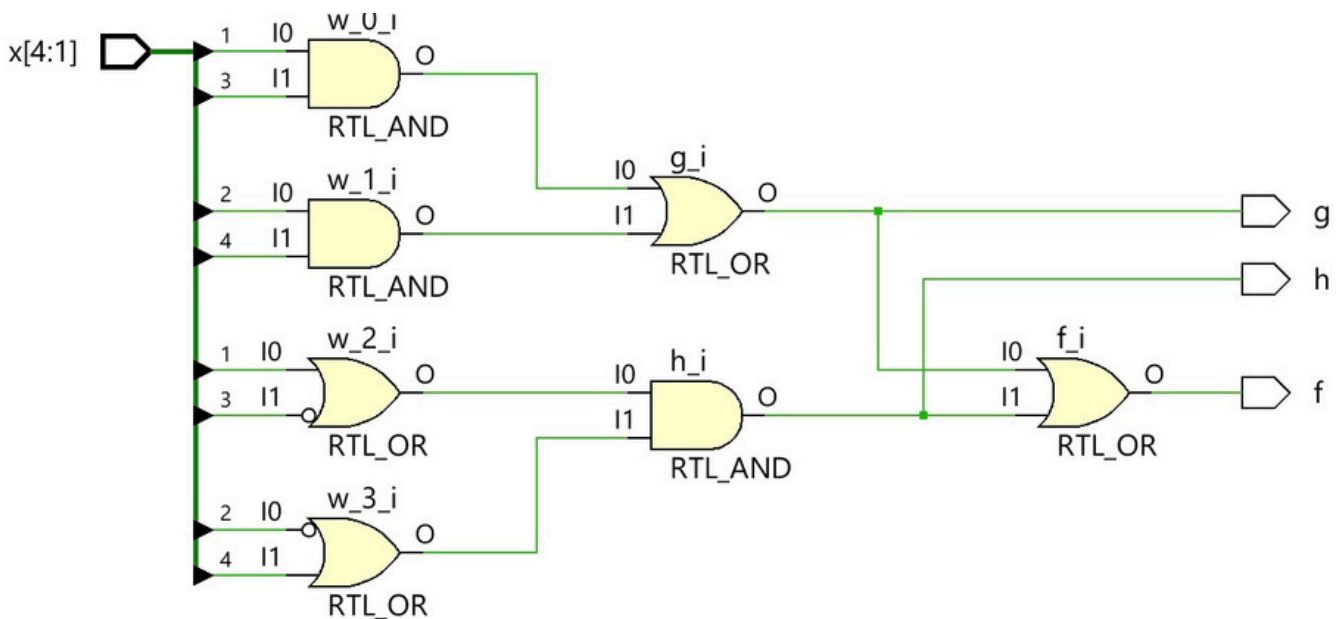
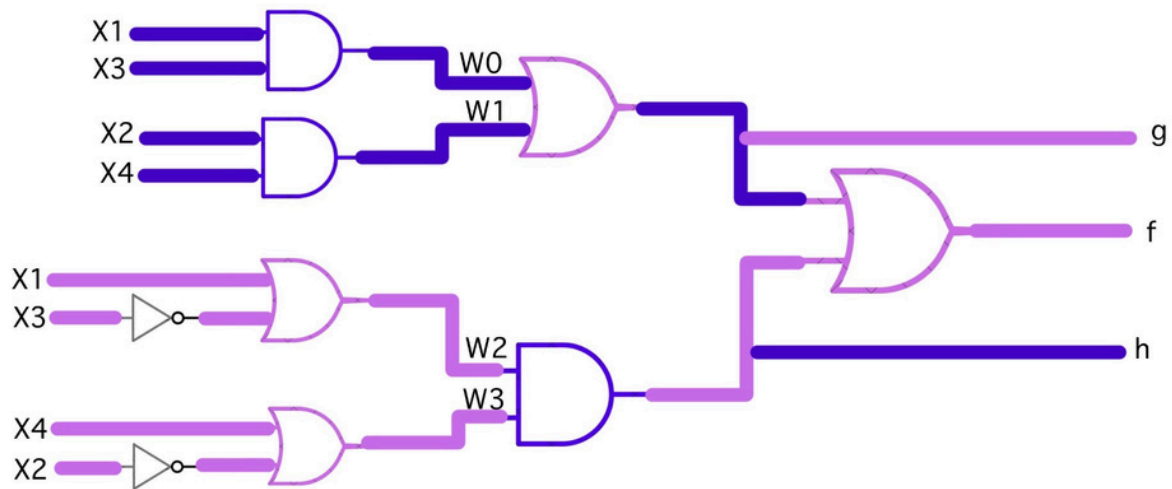
```
endmodule
```



Task #1.2

Given the following SystemVerilog code draw the microarchitecture:

```
module comb_circuit(f, g, h, x);  
  input [4:1]x;  
  output f, g, h;  
  wire [3:0]w;  
  and (w[0], x[1], x[3]);  
  and (w[1], x[2], x[4]);  
  or (w[2], x[1], ~x[3]);  
  or (w[3], ~x[2], x[4]);  
  or (g, w[0], w[1]);  
  and (h, w[2], w[3]);  
  or (f, g, h);  
endmodule
```

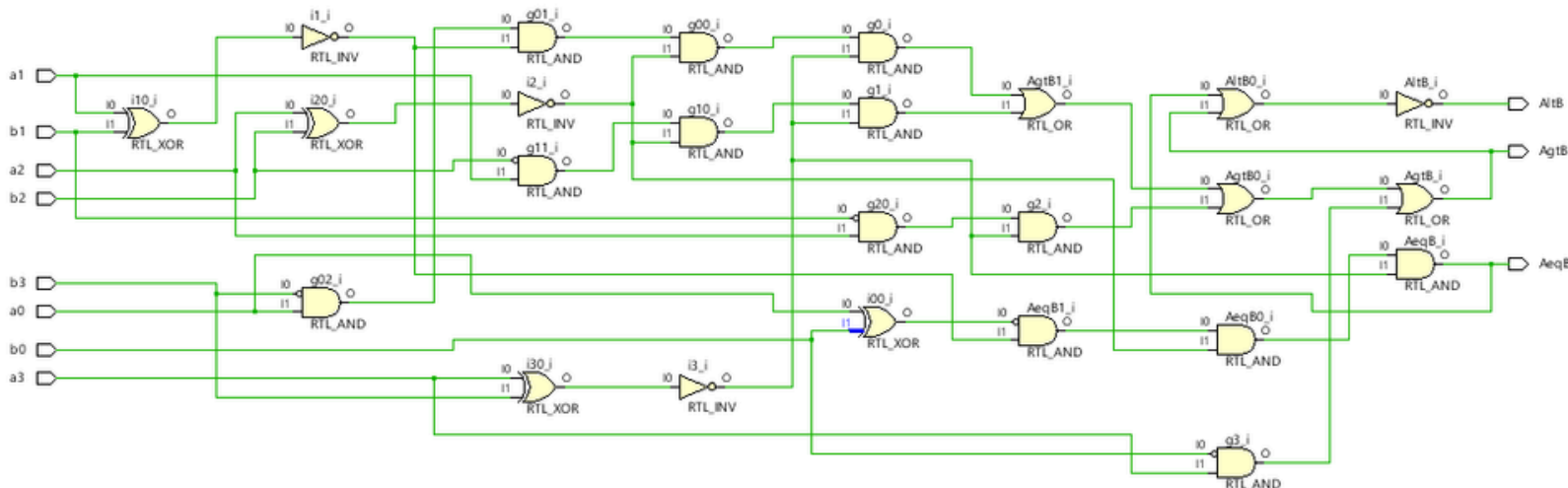


Task #1.3

```

module task3(
    input logic a0,
    input logic b0,
    input logic a1,
    input logic b1,
    input logic a2,
    input logic b2,
    input logic a3,
    input logic b3,
    output logic AeqB,
    output logic AltB,
    output logic AgtB);
    logic i3,i2,i1,i0;
    logic g0,g1,g2,g3;
    logic nb0,nb1,nb2,nb3;
    xnor (i3,a3,b3);
    xnor (i2,a2,b2);
    xnor (i1,a1,b1);
    xnor (i0,a0,b0);
    not n0 (nb0,b0);
    not n1 (nb1,b1);
    not n2 (nb2,b2);
    not n3 (nb3,b3);
    and (AeqB,i0,i1,i2,i3);
    and (g3,nb0,a3);
    and (g2,nb1,a2,i3);
    and (g1,nb2,a1,i2,i3);
    and (g0,nb3,a0,i1,i2,i3);
    or(AgtB,g0,g1,g2,g3);
    nor(AltB,AeqB,AgtB);
endmodule

```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task1.1	✓	✓	✓	✓	✓
Task1.2	✓	✓	✓	✓	✓
Task1.3	✓	✓	✓	✓	✓

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